

Expense Tracker Project Report

1. Introduction

Expense Tracker is a Python-based application designed to help users track and analyze their daily spending. This project demonstrates fundamental programming concepts, including user input, data storage, calculations, and data visualization. It is a simple yet practical project that can be easily understood, implemented, and maintained.

2. Project Objectives

The main objective of this project is to help users monitor their expenses and gain a better understanding of their spending habits. By organizing expenses into categories and visualizing the data, users can quickly identify where most of their money is spent.

3. Development Tools and Libraries

The project was developed using the following tools and libraries:

1. Python 3

2. Matplotlib

Python is used as the primary programming language, while Matplotlib is used to create visual charts for expense analysis.

4. Workflow

The workflow of the Expense Tracker project is shown below:

Input Expense Data → Save Expense Records → Calculate Total Expenses → Identify the Highest Expense Category → Generate Charts → Display Results

Workflow Description

1. The user enters expense categories and corresponding expense amounts.
2. The program stores the information in a dictionary.
3. The total amount of expenses is calculated automatically.
4. The program determines the category with the highest expense.
5. A pie chart and a bar chart are generated using Matplotlib.
6. The expense summary and visual charts are displayed to the user.

This workflow allows users to record, analyze, and visualize their spending patterns efficiently.

5. Program Implementation

The program begins by asking the user to enter the number of expense categories. For each category, the user provides a category name and an expense amount.

The entered data is stored in a dictionary structure. After all data has been collected, the program performs the following operations:

1. Calculates the total expense amount.
2. Identifies the category with the highest expense.
3. Displays a summary of all expense records.
4. Generates a pie chart showing the percentage distribution of expenses.
5. Generates a bar chart comparing expenses across categories.

The charts are automatically saved as image files for future reference and project documentation.

6. Test Cases

Test Case 1

Input:

Food = 120

Transport = 50

Entertainment = 80

Shopping = 150

Others = 40

Output:

1. Total Expense = 440

2. Highest Expense Category = Shopping

3. Highest Expense Amount = 150

Test Case 2

Input:

Food = 200

Transport = 100

Shopping = 180

Others = 20

Output:

1. Total Expense = 500

2. Highest Expense Category = Food

3. Highest Expense Amount = 200

Test Case 3

Input:

Food = 300

Transport = 80

Entertainment = 120

Shopping = 250

Others = 100

Output:

1. Total Expense = 850

2. Highest Expense Category = Food

3. Highest Expense Amount = 300

7. Project Results and Analysis

The project successfully records expense information, calculates spending statistics, and generates visual reports.

The generated output files include:

1. expense_pie_chart.png

2. expense_bar_chart.png

The pie chart clearly illustrates the percentage distribution of expenses among categories, while the bar chart provides a direct comparison of spending amounts. These visualizations help users better understand their spending behavior and identify areas where expenses are highest.

8. Importance of README.md

README.md is one of the most important files in an open source project because it provides essential information about the project.

Its main functions include:

1. Explaining the purpose of the project.
2. Describing the project features.
3. Providing installation instructions.
4. Providing usage instructions.
5. Explaining project requirements.
6. Presenting license information.

A well-written README file helps users and contributors quickly understand and use the project.

. Importance of requirements.txt

The requirements.txt file contains a list of external libraries required for the project.

The advantages of requirements.txt include:

1. Simplifying dependency installation.
2. Ensuring a consistent development environment.
3. Improving project reproducibility.
4. Supporting collaboration among developers.
5. Reducing installation errors.

Users can install all required dependencies using the following command:

```
pip install -r requirements.txt
```

10. Conclusion

The Expense Tracker project demonstrates how Python can be applied to solve a practical real-world problem. By recording expenses, calculating spending statistics, and generating visual reports, the application helps users better understand their financial activities.

The project also demonstrates important open source software practices, including project documentation, dependency management, licensing, and code organization. In future versions, the project can be enhanced by adding features such as data storage, monthly expense reports, and graphical user interfaces.

Overall, the Expense Tracker project is simple, practical, and educational. It provides an excellent example of how open source software principles can be combined with Python programming to create useful applications.